Supplemental Material: Shape versus circuit: why graphene geometric diodes and Tesla valves do not make terahertz rectennas

Todd Chapman

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This supplement lists the acceptance tests that each solver passed before it was used, the closing convergence checks, and the numerical parameters of the production runs. Every test is a program in the public repository and prints the measured number against its target; the test names below are those of the code, with the plasmonic tests prefixed P to distinguish them from the ballistic ones.

S1. BALLISTIC SOLVER

TABLE S1. Acceptance tests of the trajectory solver at dc (200 000 particles per run). All pass.

Test	What is checked	Target
T1	specular straight channel, no field	transmission exactly 1
T2	diffuse-wall channel, length to width 0.5 to 16	two-dimensional free-molecular transmission factors [1] 0.804, 0.683, 0.541, 0.400, 0.274, 0.179
T3	asymmetric funnel, no field, $p = 1$ and 0.3, $\lambda = 700\,\mathrm{nm}$	$W_S T_{SD} = W_D T_{DS}$ within 3.5 standard errors
T4	the same funnel with the field on, both injection sides at one total energy	$N_S T_{SD} = N_D T_{DS}$
T5	mirror-symmetric constriction under bias	asymmetry 1
T6	zero bias	zero injection weight and zero current
T7	three-stage side-loop valve, diffuse walls	trapped, leaked and Dirac-cut fraction below 10^{-3}
T8	ballistic $1\,\mu$m channel at 10^{12} cm^{-2}	Sharvin conductance 8.74 mS within 1 %
T9	$10\,\mu$m channel, $\lambda = 300\,\mathrm{nm}$	$T = \ell_1/(L + \ell_1)$, $\ell_1 = \pi\lambda/2$, within 5 %
T10	chord tolerance 0.05 against 0.005 nm	same transmission
T11	funnel of 100/10/40 nm (lead, neck, drain) at zero field	$T = 0.0995$ against the neck-to-lead ratio 0.100 (information)

TABLE S2. Acceptance tests of the time-dependent solver and the two-sided estimator (10^6 particles per side; T16 4×10^6, T17 2×10^6). Currents in μA for the clean specular funnel at 0.1 V. All pass.

Test	What is checked	Result
T12	integration of $\hbar\dot{k} = F$ without energy re-imposition, static field, against the dc solver	$T(+0.1) = 0.14141$ (reference 0.14130), $T(-0.1) = 0.04148$, $A = 1.622$ (reference 1.624)
T13	two-sided equilibrium estimator against the bias-window estimator; zero bias	188.7 ± 0.6 against 189.2 ± 0.5; -0.15 ± 0.66 at $V = 0$
T14	quasi-static anchor at 300 MHz: rectified current and $\lvert Y_1 \rvert$ against the static curve	17.78 ± 0.59 against 18.27 ± 0.18; 1.589 ± 0.013 against 1.580 mS
T15	straight control at 3 GHz and 1 THz	-0.80 ± 0.57 and -0.20 ± 0.62; trapped plus leaked below 2×10^{-6}
T16	chord cap $v_F/40f$ against $v_F/160f$, 500 nm drop, 1 THz	6.32 ± 0.30 against 5.55 ± 0.32
T17	Dirac cut against a classical turning point, 500 nm drop, 1 THz	5.94 ± 0.45 against 5.69 ± 0.45

The quasi-static anchor is at 300 MHz and not at a few gigahertz because of the slow-drift population described in the main text. The equilibrium injection of the two-sided estimator samples the kinetic distribution down to zero energy, and between 10^{-3} and 7×10^{-3} of the injected electrons are driven below the 2 meV cut that stands in for the Dirac point; these would return to their own contact and carry no current. In the harmonic estimator a drain-injected electron removed by the cut must be booked as returned, otherwise $\lvert Y_1 \rvert$ at the anchor is 6% low. Five operating points

recomputed with a classical turning point in place of the cut agree with the production values within one standard error.

S2. VISCOUS SOLVER

TABLE S3. Acceptance tests of the steady Navier-Stokes solver. All pass.

Test	What is checked	Result
H1	Poiseuille flow, no slip and Navier slip $S = 0.5$, 2, ∞	exact to 10^{-12}
H2	Brinkman-Poiseuille flow, Da = 10, $S = 0.5$; Brinkman plug flow	velocity error $O(h^{2.7})$, 1.2×10^{-6} at $h = 25\,\mathrm{nm}$; plug flow exact
H3	Stokes reciprocity of a side-loop valve, three (Da, S)	$D - 1 = 2 \times 10^{-12}$ to 2×10^{-11}; wall flux exactly zero
H4	Kovasznay flow [2], Re = 40	velocity error $O(h^{2.9})$, 2.2×10^{-5} at $h = 25\,\mathrm{nm}$
H5	backward-facing step [3, 4], expansion ratio 2	reattachment at 2.90 step heights (Re = 100) and 4.96 (Re = 200)
H6	side-loop valve as a classical no-slip device	$D = 0.9993$, 1.0037, 1.0197 at Re = 25, 50, 100
H7	mesh convergence, side-loop valve, Re = 10	$D - 1 = 1.01$, 1.10, 1.14×10^{-2} at wall spacing 20, 14, 10 nm
H8	mirror-symmetric constriction, Re = 10	$D - 1 = 7 \times 10^{-6}$
H9	$(D-1)/\mathrm{Re}$ at Re = 0.05 to 0.2; mirrored and re-meshed device	constant to 2 %; directional solver bias 6×10^{-7}
U1	units	Poiseuille and Brinkman resistances equal the Gurzhi and Drude formulas to 10^{-12}

Below $\mathrm{Re} \approx 1$ the diodicity is under the discretisation floor of about 3×10^{-5} and is reported as a bound. For the classical valve with no-slip walls the steady branch continued from Stokes flow ends at a fold near Re = 85; this is outside the range relevant to graphene.

TABLE S4. Mesh convergence of the classical Tesla valve (175 nm channel, no damping): $D - 1$ at wall spacings of 20 nm (16 703 nodes, the production mesh), 14 nm (25 447 nodes) and 10 nm (38 449 nodes).

Wall condition	Re	20 nm	14 nm	10 nm
perfect slip	3	0.0119	0.0118	0.0124
perfect slip	10	0.234	0.232	0.243
perfect slip	30	1.472	1.510	1.619
slip length $0.5\,W$	3	0.0068	0.0068	0.0072
slip length $0.5\,W$	10	0.151	0.151	0.159
slip length $0.5\,W$	30	1.145	1.177	1.257

Table S4 shows that the diodicity of the classical valve changes by less than 3% between the two coarser meshes and by 5 to 7% between the two finer ones; the pressure drop itself converges only at first order because of the re-entrant corners at the loop junctions. The values quoted in the main text are from the 20 nm mesh and are good to about 10%.

S3. PLASMONIC SOLVER

The time-domain runs start from the perturbative state of the time-discretised operator, which removes a start-up transient that would otherwise exceed the rectified signal at small amplitude; the dc load current is held at the small-signal prediction and corrected with the dc Thevenin relation. The level of the square law carries the 60-step time-discretisation bias (+0.6%, +4.5% and −0.5% at 0.1, 0.3 and 1 THz) and the 2% mesh uncertainty of P8; the ratios to the square law quoted in the main text are taken against the time-discrete law, so the bias cancels. The shape of the response is converged in the time step: at 100 K the efficiency ratio between 60 and 120 steps per period is 0.991 (P12), and at 300 K the ratio of the efficiency to the square law changes from 0.992, 1.054, 1.101 to 1.004, 1.058, 1.109 at amplitudes 0.1, 0.5, 0.7 of the voltage unit at 0.3 THz, and from 1.144 to 1.145 at 0.7 at 0.1 THz.

TABLE S5. Acceptance tests of the perturbation solver (P1 to P9, P15 to P17) and of the time-domain solver (P10 to P14). All pass.

Test	What is checked	Result
P1	static limit against the viscous solver	relative difference 5×10^{-4}
P2	linear plasmon response of a channel against $1/\cos kL$	errors 7×10^{-9} to 1.2×10^{-5}
P3	position and width of the first plasmon resonance, with friction, viscosity and viscoelastic viscosity	position 0.1568 (exact 0.1571); widths within 1 %
P4	photoresponse against the Dyakonov-Shur closed form [5] and an independent quadrature	errors 2×10^{-7} to 2×10^{-4}
P5	funnel, quasi-static rectified current: curvature of the steady solution against perturbation	2.206×10^{-2} against 2.210×10^{-2}
P6	time domain against perturbation at small amplitude	ratios 0.9997 and 0.9994
P7	symmetric termination: dc current of a symmetric constriction and of a channel relative to the funnel	6×10^{-7} and 10^{-8} against 1.6×10^{-3}
P8	mesh convergence of the funnel response on three meshes	last change 2.0 % and 1.1 % at reduced frequency 1 and 3
P9	units	plasma speed, gate overdrive, resistance and frequency units against hand values
P10	time-domain efficiency against the time-discrete perturbation law	ratios 1.0033 and 1.0038
P11	dc load correction from a deliberately wrong bias seed	1.0016 (short circuit), 1.0017 (open circuit)
P12	time step: 60 against 120 steps per period at 100 K	efficiency ratio 0.991
P13	power bookkeeping	absorbed power within 0.2 % of the linear value
P14	P10 for the symmetric termination	ratio 0.998
P15	contact resistance: $I_{\mathrm{sc}}(R_{\mathrm{dc}} + 2R_c) = V_{\mathrm{oc}}$; port relations; passivity	7×10^{-15}; 4×10^{-16}; absorbed fraction 0.39
P16	invariance under co-scaling of density, gate distance, width and scattering times at fixed nondimensional groups	nondimensional response identical to round-off
P17	units and closure at a second operating point	closed forms to 2×10^{-16}; Dirac against shallow-water response ratio 0.970

S4. PRODUCTION PARAMETERS

Ballistic dc: 400 000 trajectories per bias point in 64 batches. Ballistic ac: 10^6 trajectories per contact and point (4×10^6 at $\lambda = 700\,\mathrm{nm}$ and at the smaller amplitude), chords limited to 40 nm, to $v_F/40f$ and to one twentieth of the field region. Viscous: Taylor-Hood meshes with 20 nm wall spacing (23 000 to 55 000 triangles), Newton residual below 10^{-10}. Plasmonic small signal: wall spacing 14 nm for the 175 nm devices and 80 nm (18 601 nodes) for the 1 μm funnel, 25 frequencies from 10 GHz to 3 THz. Plasmonic large signal: the same 1 μm mesh, 60 steps per period, eight amplitudes from 0.01 to 0.7 of the voltage unit.

[1] E. A. Richley and T. W. Reynolds, *Numerical solutions of free-molecule flow in converging and diverging tubes and slots*, NASA Technical Note TN D-2330 (National Aeronautics and Space Administration, Lewis Research Center, Washington, D.C., 1964).

[2] L. I. G. Kovasznay, Laminar flow behind a two-dimensional grid, Mathematical Proceedings of the Cambridge Philosophical Society **44**, 58 (1948).

[3] B. F. Armaly, F. Durst, J. C. F. Pereira, and B. Schönung, Experimental and theoretical investigation of backward-facing step flow, Journal of Fluid Mechanics **127**, 473 (1983).

[4] E. Erturk, Numerical solutions of 2-D steady incompressible flow over a backward-facing step, Part I: High Reynolds number solutions, Computers & Fluids **37**, 633 (2008).

[5] M. Dyakonov and M. Shur, Detection, mixing, and frequency multiplication of terahertz radiation by two-dimensional electronic fluid, IEEE Transactions on Electron Devices **43**, 380 (1996).